

Statistical Mechanics of Money:

Applying Boltzmann-Gibbs Distributions and Entropy
to Economic Dynamics via Agent-Based Simulation

Submitted for:

Physics of Complex Systems (Exam)

Università della Magistrale in Fisica

September 1, 2026

Contents

Abstract

This paper investigates the applicability of **statistical mechanics** to economic systems, specifically how money distribution in a society of interacting economic agents follows the **Boltzmann-Gibbs distribution**—a foundational principle from thermodynamics and statistical physics.

We conducted agent-based simulations modeling economic transactions as random exchanges of money between agents, analogous to particle collisions in a gas. Under various transaction mechanisms (constant exchanges, fraction of individual wealth, fraction of system average), the simulated money distribution converges to an exponential form $P(m) = \frac{1}{T}e^{-m/T}$, where $T = M/N$ represents the system’s effective “temperature” (average wealth per agent).

Three simulation models are analyzed:

1. **Basic money distribution** with no debt constraints
2. **System with debt** allowing limited negative balances
3. **Stock market extension** incorporating both cash and equity trades with price dynamics

We demonstrate that **Gibbs entropy** correctly predicts equilibrium states and that the system evolves as a **Markov process** toward maximum entropy. These results validate the hypothesis that economic inequality and wealth concentration emerge as natural equilibrium states when agents interact via random transactions, analogous to how physical systems spontaneously approach thermodynamic equilibrium.

This paper connects fundamental concepts from statistical mechanics—Markov processes, Boltzmann-Gibbs distributions, Gibbs entropy, and dynamical systems theory—to economics, providing a computational validation of theoretical predictions.

Keywords: Statistical mechanics, Boltzmann-Gibbs distribution, Gibbs entropy, Agent-based simulation, Markov processes, Economic dynamics, Complex systems

1 Introduction

1.1 Motivation

[Introduction content to be filled with detailed physics concepts...]

2 Model and Theory

2.1 Theoretical Framework

[Model and theory content to be filled...]

3 Results

3.1 Scenario 1: Basic Money Distribution

[Results content to be filled with figures and analysis...]

4 Conclusion

4.1 Summary

[Conclusion content to be filled...]

A Mathematical Details

A.1 Boltzmann-Gibbs Distribution

[Mathematical details to be filled...]

B Code Snippets

B.1 Core Simulation

[Code snippets to be filled...]